ELI — The Complete User Manual

ELI v2.3.15

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Contents

ELI — The Complete User Manual

Plain-English Edition · with flowcharts

A friendly, no-jargon guide to everything ELI can do and how to make it do it.

This manual assumes **zero** technical background. You talk to ELI in normal English — by **typing** in the chat box or **speaking** — and it does things on your computer, answers questions, remembers what matters to you, and learns your routines. Everything runs **on your own machine**; nothing is sent to the cloud.

> **The one rule worth knowing:** ELI is **offline by default**. It only touches the internet when *you* ask for something that needs it (a web search, the news, downloading a model). Your conversations, files, and memories never leave your computer.

How to read this manual: - **Just want to use it?** Read §1–§5 and the cheat sheet (§18). That's enough to be productive. - **Want the whole picture?** Read straight through — every feature is here, in plain terms. - **[Everyone] / [Some setup] / [Advanced] tags** at the start of some sections mean *anyone / a little technical / advanced*.

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1. What ELI is (in one minute)

ELI is a **personal AI assistant that lives on your computer**. Think of it as a capable helper that can:

- **Talk with you** like a knowledgeable person — by text or voice.
- **Operate your computer** — open apps, play music, manage windows, type for you.
- **Look at things** — your screen, images, PDFs, spreadsheets — and explain them.
- **Write for you** — documents, notes, code, scripts.
- **Remember** what matters about you, and **learn** your daily routines.
- **Act on its own** when helpful — a morning briefing, a heads-up, a suggestion.

The big difference from cloud assistants: **it's yours**. It runs locally, stays private, has no subscription, and you can even retrain its "voice" yourself (§14).

Here's the whole thing at a glance:

```
mindmap
  root((ELI))
    Talk
      Type or speak
      Wake word
    Do
      Open apps
      Play music
      Manage windows
      Type for you
    See
      Your screen
      Images and PDFs
      Spreadsheets
    Create
      Documents
      Notes
      Code and scripts
    Know you
      Memory of facts
      Living profile
      Learned habits
    Act on its own
      Morning briefing
      Suggestions
      Overnight tasks
    Private
      Runs locally
      Offline by default
      No accounts
```

---

2. How ELI works when you ask it something

[Everyone] *Anyone — this is the mental model that makes everything else click.*

When you say or type something, ELI follows the same honest loop every time:

```
flowchart TD
    A([You ask ELI something<br/>type or speak]) --> B{Is it a command<br/>or a conversation?}
    B -->|A command<br/>'open spotify'| C{Does it need<br/>the internet?}
    B -->|A conversation<br/>'how are you?'| H[ELI replies in its<br/>own voice]
    C -->|No| D[ELI does it<br/>on your computer]
    C -->|Yes, e.g. news/search| E{You allowed<br/>online for this?}
    E -->|Yes| F[ELI fetches it,<br/>tells you it went online]
    E -->|No| G[ELI stays offline<br/>and says so]
    D --> I{Did it actually<br/>work?}
    F --> I
    I -->|Yes| J([ELI confirms what it did])
    I -->|No / unsure| K([ELI tells you honestly —<br/>never pretends it worked])
    H --> J
```

The promise in that diagram: ELI is **built not to fake an action**. When it says it played a song or fixed a file, that claim comes from the code that actually ran it — and when it couldn't, it is designed to say so plainly instead of bluffing. This is enforced in software, not just good manners. It is a strong guard rather than a mathematical guarantee: no filter over a language model can promise perfection, so ELI is built to fail loudly rather than quietly invent.

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3. Getting started

[Some setup] *A little technical, but only once.*

Installing (v2.1.23 and later — one download per platform)

Get the latest from **GitHub Releases**. Every release is built in CI and **launch-tested on Windows, macOS and Linux** before it can publish.

- **Windows:** run `ELI-Setup-<version>.exe` — installs per-user, no admin password. If it sees an NVIDIA card it offers GPU acceleration during setup; if it finds an older ELI it offers a **fresh install** (resets settings & memory; keeps downloaded models). You get three shortcuts: **ELI**, **ELI Server**, **Uninstall ELI**.
- **macOS (Apple Silicon):** open the `.dmg`, drag **ELI** into Applications. First launch is right-click → **Open** (the app is unsigned). Apple-GPU acceleration is built in — nothing to set up.
- **Linux:** `chmod +x ELI_v2-<version>-x86_64.AppImage` and run it. On first launch it offers **applications-menu entries** (ELI, ELI Server, and a working Uninstall).

```
flowchart TD
    A([Run the installer / AppImage]) --> B[First launch]
    B --> C{GPU detected?}
    C -->|NVIDIA| D[Offer: CUDA engine<br/>one-time download, verified]
    C -->|AMD / Intel Arc| E[Offer: Vulkan engine<br/>one-time download, verified]
    C -->|Apple| F[Metal built in]
    C -->|None / decline| G[CPU engine - always works]
    D --> H{Chat model installed?}
    E --> H
    F --> H
    G --> H
    H -->|No| I[Offer: starter model<br/>sized to your hardware]
    H -->|Yes| J
    I --> J[Quick interview]
    J --> K([Ready - say hello])
```

What arrives on first launch

Asset	When	Size
Piper voices (speech out)	already inside the installer	—
Nomic embedder (memory/recall)	fetched automatically	~80 MiB
Whisper (speech in)	fetched on first voice use	~460 MiB
GPU engine (optional)	if you accept the GPU offer	~90 MiB (AMD) · ~1–2 GiB (NVIDIA)
Chat model	if you accept the model offer	sized to your hardware (e.g. ~4.7 GiB)

Everything ELI knows and downloads lives in **one per-user folder** that **survives upgrades** — like a save-game: `%LOCALAPPDATA%\ELI_v2` (Windows), `~/Library/Application Support/ELI_v2` (macOS), `~/.local/share/ELI_v2` (Linux). Databases start as a **true blank slate** — no personal data ships in any installer.

Starting over, or uninstalling

- **Clean slate:** tick *Fresh install* in the Windows setup, or run `ELI --fresh-start` anywhere (keeps downloaded models unless you add `--purge-models`).
- **Uninstall:** Windows → the **Uninstall ELI** shortcut. Linux → the **Uninstall ELI** menu entry (removes the entries, offers to delete your data, then delete the AppImage file). macOS → drag the app to Trash; your data folder is listed above.
- **GPU later / undo:** `ELI --install-gpu-pack` (add `--vulkan` for Intel Arc) · `ELI --remove-gpu-pack`.
- **From source (developers):** `bash install.sh` from a clone works exactly as before.

First launch — the quick interview

The very first time you open ELI, it doesn't know you yet, so it asks **three short questions** (your name, what you do, how you like to be spoken to). Ten seconds, and you can **skip it** entirely. From there, ELI builds its understanding of you naturally as you talk.

Opening ELI

- **Desktop app (full experience):** the **ELI** shortcut / menu entry (installer builds), the AppImage itself (Linux), or `scripts/eli_launch.sh` from a source clone.
- **From your phone:** the **ELI Server** shortcut (or `ELI --server`) starts the home server with the connect QR — see §13.

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4. Your first hour with ELI

[Everyone] A gentle on-ramp. Try these in order — each builds confidence.

1. **Say hello.** Type `how are you?` — get a feel for ELI's voice.
2. **Ask what it can do.** `what can you do?` lists its capabilities.
3. **Open something.** `open the file manager` or `open firefox`.
4. **Play music.** `play lo-fi on YouTube`, then `pause`, then `skip`.
5. **Look at your screen.** `what's on my screen?`
6. **Tell it a fact.** `remember that my dog's name is Rufus`. Later: `what's my dog's name?`
7. **Set a reminder.** `set a timer for 5 minutes`.
8. **Get a briefing.** `what's the news?` (this one goes online — ELI will say so).
9. **Check on ELI itself.** `what are you running on?` shows its model and hardware.
10. **Go hands-free.** Say the wake word **"computer"**, then a request.

By the end you've touched conversation, control, vision, memory, reminders, the web, voice, and introspection — the whole surface.

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5. Talking to ELI

There are two ways, freely mixed:

- **Type** in the chat box and press Enter.
- **Speak** — click the microphone, or say the **wake word** (default **"computer"**) then your request. Change the wake word to anything (§11).

You don't need special phrasing. Talk normally:

“open spotify and play some lo-fi” “what’s on my screen?” “remember that my sister’s name is Anna” “set an alarm for 7am”

Chaining: ask for several things at once — > “close steam and set an alarm for 7am” > “open spotify then play Juicy by Notorious B.I.G.”

If ELI isn’t sure what you meant, it asks — it won’t guess wildly or pretend. And if a request needs the internet, it tells you before going online.

If you’re having a hard time: ELI is tuned to notice genuine first-person distress (it’s built to ignore ambient audio and dark jokes, not to over-react). When it does, it steps out of task-mode and gently checks in — pointing you toward real human support rather than ploughing on with commands or reciting a canned line. It’s a steer, not a script.

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6. A tour of the window

[Everyone] ELI Pro is organised into **tabs**. You’ll mostly live in **Chat**; here’s the rest in plain terms:

Tab	What it’s for
Chat	Talk to ELI, by voice or text. Home base.
Quick Actions	One-tap buttons for the things you do most.
Memory	Browse what ELI remembers about you; add or remove facts.
Habits	Routines ELI has learned; turn on/off, edit times, add your own.
Proactive	Controls for ELI’s “act on its own” behaviour (suggestions, summaries, insights).
Tasks	Scheduled & overnight jobs and their status.
Images	Generate images — procedural (fast) or diffusion (photoreal).
Screen	Watch/read your screen, take screenshots, OCR.
Coding / IDE	Write, examine, and fix code; a built-in editor.
Self-Improve	What ELI is proposing to improve in *itself* — approval-gated.
Report Builder	Generate longer documents/reports with sources (evidence → outline → draft → revise).
Elis World	ELI’s own world-model view.
Labs	Power-user workshop — notebook, Jupyter, calculator, physics, file-chat, **projects/workspaces**, sim-IDE, orchestration, test review, **training** (§14).
Settings	Model (incl. the model-switch dropdown), hardware, voice, privacy options (§16), plugins and the **Marketplace** (§20).

You rarely need the tabs directly — almost everything is reachable by **asking** in Chat: “show my habits”, “what do you remember about me”, “show background jobs”, “make an image of …”, and ELI opens or reports the right thing.

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7. What ELI can do — by everyday task

[Everyone] Real things you can say. ELI understands many phrasings — these are examples.

Music & media

- "play lo-fi on YouTube" · "play Juicy by Notorious B.I.G. on Spotify"
- "pause" · "skip" · "next song" · "previous track" · "shuffle" · "repeat this song"
- "what's playing?" · "stop the music" · "skip the ad" (YouTube, when skippable)

Apps & windows

- "open spotify" · "launch firefox" · "open the file manager" · "open github.com"
- "close chrome" · "focus the browser" · "switch to workspace 2"
- "tile windows" · "maximise the window" · "show desktop" · "next window"

Volume, typing, mouse

- "volume up" · "mute" · "set volume to 30"
- "type hello world" · "press enter" · "left click" · "move the mouse up"

Files & notes

- "create a file notes.txt" · "make a folder called projects" · "read notes.txt"
- "list the files in ~/Documents" · "write a note saying buy milk" · "search notes for budget"
- "what's in my clipboard?" · "summarise report.docx" · "convert report.md to pdf"

Writing, documents & code

- "write a report on renewable energy" · "generate a document about X"
- "write a bash script to monitor the GPU" · "solve this: implement a function that …"
- "fix the bugs in foo.py" · "examine eli/memory/memory.py for errors" → ELI scans, **offers** fixes, and applies them only if you say "yes, fix it". · "show the diff"
- "fabricate a test dataset for X" → ELI generates synthetic/test data on a topic.
- The **Report Builder** tab turns "generate a document on X" into a real report: it gathers evidence, plans an outline, drafts each section, then **reviews and revises** it — with document-type quality profiles. It's generation-first, not a one-shot dump.

Screen, images, PDFs, spreadsheets

- "what's on my screen?" · "find the submit button on screen" · "take a screenshot"
- "describe cat.jpg" · "read the text in image.jpg" (OCR)
- "summarise report.pdf" · "analyse data.csv" · "watch my screen" / "stop watching"

Information

- "what time is it?" · "what's the date?" · "weather in Wexford"
- "what's the news?" / "catch me up" → a **synthesised** briefing, not a raw dump
- "search the web for X" (goes online — ELI tells you) · "gpu status" · "system stats"

Reminders, timers, calendar, scheduling

- "set an alarm for 7am" · "set a timer for 10 minutes"
- "add an event tomorrow at 3pm" · "list my events" · "start a pomodoro"
- "research the best solar inverters overnight" · "build me a script at 2am" → **background jobs**
- "show background jobs" · "check job 5"

Asking ELI about itself (honest introspection)

- "what are you running on?" (model, context, GPU) · "how does your memory work?"
- "what do you know about me?" · "what updates have you made recently?" · "status"

Eye control (gaze)

- “calibrate gaze”, then “enable gaze”. With it on, ELI clicks **where you’re looking** when you say “open” / “left click” / “right click” / “hit enter” — hands-free pointing.
- **Fully hands-free *and* voice-free — dwell-click.** Say **“enable gaze clicking”** (or turn on **Dwell-click** in Settings) and ELI clicks wherever you **rest your gaze for about a second** — no mouse, no voice needed. This is built for people who can’t use their hands *or* speak reliably. Tune the hold time, wander radius, and sensitivity in Settings (`gaze_dwell_seconds`, `gaze_dwell_radius`). Turn it off any time; it’s opt-in.
- “gaze status” · “disable gaze”

Making images

- “make an image of a red bike by the sea” · “generate a picture of …”
- Two engines: a fast **procedural** one (always available) and **diffusion** (SSD-1B) for photoreal results — ELI briefly frees the model’s VRAM to run it. The **Images** tab has the controls; results save locally.

Coding & building things

- “solve this: implement a function that …” → ELI plans it, writes it, then **verifies and repairs** it.
- “generate a project that does X” (a full multi-file scaffold) · “write a bash script to …”
- “examine eli/memory/memory.py for errors” → a tiered scan; it **offers** fixes and applies them only if you say “yes”. · “show the diff” · “generate tests for your code” (it writes *and* sandbox-verifies them). The **Coding** tab and the built-in **IDE** are home for this.

ELI improving itself

- “improve yourself” → a self-improvement patch cycle — proposed, and **gated by your approval**.
- “show your self-improvement log” · “patch yourself” · “upgrade yourself” (git pull → deps → rebuild indexes) · “run your self-tests”. The **Self-Improve** tab shows what it’s proposing.

Plugins & your own agents

- “list plugins” · “enable the X plugin” · “install the X plugin” · “plugin status”
- You can **build your own agent** just by describing it — ELI runs a short dialog, validates it, and registers it live (the create-agent flow).

Overnight & scheduled work

- “research the best solar inverters overnight” · “build me a script at 2am” · “every night, regenerate the test report” → durable **background jobs** that survive restarts.
- “show background jobs” · “check job 5”. The **Tasks** tab lists them.

Shaping ELI’s voice (persona)

ELI’s personality is **emergent** — it forms from your profile, your history, and the way you talk to it, not from a hand-written script. Most of the time you leave it alone and it just *becomes* itself with you. When you want to steer it, you can, without erasing that: - “lock your persona to a terse senior engineer” · “what persona are you locked to?” · “clear persona lock”. A lock **layers on top** of the emergent voice for the session; clearing it hands the wheel back to the natural persona. - “refresh your persona” reloads it from your latest profile (after ELI’s learned something new about you). - **Set how it talks to you** *(new, 2.1.23)* — say *“speak to me more bluntly”*, *“be funnier”*, or *“respond as Rick”* and it **sticks** (say *“reset your tone”* to clear it). You can also set it in **Settings → Identity → Tone / how ELI speaks**. This explicit preference takes priority over the tone ELI infers, but leaving it blank keeps ELI’s natural emergent voice in charge. - ELI also keeps a small set of **core values** (memory policy, verbosity, safety) that are part of its persona and evolve alongside it.

The rule of thumb: **the emergent voice is the default and the truth of who ELI is with you; a lock is a temporary hat it wears on request.** ELI won't drift into a scripted character on its own, and a lock never overwrites the underlying personality — it sits on top until you clear it.

Tone & emotion — reading the room

ELI now **reads the emotional situation from two independent signals** — the *sound* of your voice (energy, mood) and the *content* of what you say — and shades **how** it delivers: its wording, its voice (pace, warmth, pauses), and its avatar expression all move together. If you're clearly frustrated it gets calmer and more direct; if you're playing, it plays back; if you're upset, it softens. **This never changes who ELI is** — its core personality stays exactly the same; it just picks the tone that's most useful for the moment.

You can also set the register yourself: *"be comedic"*, *"talk street"*, *"sound more professional"*, *"use a deadpan tone"* — and *"go back to normal"* (or *"be yourself"*) to hand it back to ELI's own read of the room. There's a growing palette of tones (comedic, professional, curious, deadpan, tender, street-smart, gremlin, and more), and you can add your own in `config/voices/tones.json`. Turn the whole thing off with `ELI_TONE_ADAPT=0`.

---

8. ELI remembers you (memory)

[Everyone] ELI builds a private, evolving understanding of **who you are** and reads it every time you talk, so you never repeat yourself.

```
flowchart TD
    A([You're talking to ELI]) --> B{What kind of thing<br/>did you say?}
    B -->|'Remember that ...'| C[Saved as a durable fact<br/>kept until you remove it]
    B -->|Just chatting<br/>interests, focus, tone| D[Quietly updates ELI's<br/>living profile of you]
    C --> E[(Local memory<br/>on your computer)]
    D --> E
    E --> F([Read every turn so ELI<br/>stays in context])
    G[/Old, abandoned topics<br/>fade over time/] --> E
```

Tell it something to keep: "remember that my sister's name is Anna" · "my name is Alex" **Ask what it knows:** "what do you remember about my car?" · "what do you know about me?" · "explain everything you know about me and where it's stored" (a sourced, detailed answer).

It remembers the thread, not just facts: when a conversation ends, ELI quietly summarises it and carries a digest of your recent sessions forward — so it recalls what you've been *working through* over days, not just isolated one-off facts.

Everything is stored **locally** in small private databases. Review and prune it in the **Memory** tab. A brand-new install is a **blank slate** — ELI knows nothing about you until you talk.

---

9. ELI learns your routines (habits)

[Everyone] If you do the same thing at roughly the same time on **several different days**, ELI notices and **offers** to automate it. It never activates anything without your "yes".

```
flowchart TD
    A[You open your email app<br/>around 9am, several days] --> B{Same action, same time,<br/>on 3+ differe
    B -->|No| X[ELI stays quiet<br/>no nonsense suggestions]
    B -->|Yes| C[ELI offers:<br/>'Add this as a habit? yes/no']
    C --> D{Your answer}
    D -->|No| E[Dropped, won't ask again]
```

```
    D -->|Yes| F{Has today's time<br/>already passed?}
    F -->|Yes| G[Runs it now once,<br/>then daily from tomorrow]
    F -->|No| H[Runs at the time, daily]
    G --> I([Active habit — edit in Habits tab])
    H --> I
```

Manage everything in the **Habits** tab (add, edit times, turn off), or say "show my habits". The key safeguards: it only proposes routines you've **genuinely repeated across days**, and it **always asks first**.

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10. ELI helps before you ask (proactive)

[Everyone] With **proactive mode** on, ELI can do helpful things on its own — a **morning briefing**, a gentle suggestion, or a goal it thinks would help. It's conservative and easy to control:

- "start proactive mode" / "stop proactive mode" · "proactive status"
- "morning report" / "give me my briefing" (news, weather, your agenda)
- "do you have any proposals for me?" / "what's on your agenda?"

Schedule real unattended work too: > "research X overnight" · "generate tests for your code tonight"

These appear in the **Tasks** tab and survive restarts.

The autonomy tick — what "on its own" actually means

When you allow it, ELI runs a quiet **autonomy tick** inside the proactive daemon — roughly every half hour, and only if it's switched on (kill switch: `ELI_AUTONOMY_TICK=0`). One tick does three things, all **read-only or proposal-only**:

1. **Code monitor** — it glances over its own code health and flags anything that's drifted.
2. **Self-model refresh** — it updates its understanding of itself (what it's running, what it can do) so its introspection stays honest.
3. **Goal autogenesis** — it may propose a goal, or a scheduled task, it reckons would help you.

The guardrail that matters: every autogenerated goal is **proposal-only**. It shows up as a suggestion you accept or dismiss — nothing destructive ever runs unattended, and anything risky still goes through the **approval gate** (Appendix C). Autonomy makes ELI *thoughtful* on its own, not *loose*.

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11. Voice, wake word, and dictation

[Everyone] ELI is fully usable hands-free.

```
flowchart LR
    A([Say the wake word<br/>'computer']) --> B[ELI starts listening]
    B --> C[You speak your request]
    C --> D[ELI transcribes it<br/>on your computer]
    D --> E[ELI does the task]
    E --> F([Optionally speaks the answer back])
```

- **Wake word:** say **"computer"** then your request. Change it: "change the wake word to athena". ELI trains its listener on the new word automatically — works even over background music.
- **Personalise to your voice:** "enroll my wake word" (records you saying it and adapts).
- **Dictation:** "start dictation" / "stop dictating" — speak and ELI types into the active field.
- **Transcribe a recording:** "transcribe audio.wav"
- **ELI speaking back:** "say good morning"

- **A human-like voice:** at the **startup model picker** (before ELI loads its model) pick **Voice engine → "Natural neural (XTTS-v2)"** for an expressive, human-sounding voice — or say *"use the natural voice"* any time. It needs a one-time `pip install "eli-v2.0[natural]"` (downloads ~1.8 GB on first use) and a reasonable GPU; on a lighter machine leave it on **Piper**, which is fast and now noticeably more natural (it pauses on full stops and lifts on questions).
- **Change ELI's voice:** the voice dropdown next to the chat box, **Settings ▸ Runtime ▸ "VOICE / TTS"**, or just ask — *"use the alan voice"*, *"switch your voice to amy"*, *"use a natural male voice"*.
- **Get more voices and accents:** ask ELI directly — *"what voices do you have"*, *"download a British voice"*, *"get me a Scottish accent"* — or use **Settings ▸ Runtime ▸ "VOICE / TTS" > "Get more voices / accents…"**. Either way, ELI can download from a library of **166 voices across 45 languages** — 38 of them English (US, British, Scottish, Northern and Southern English). Each is verified, installed, and selectable immediately. A few voices are marked *personal use only*: their licences don't allow ELI to bundle them, but you can still download them yourself.
- **Character voices:** choose `char:hal`, `char:tars`, `char:rick`, `char:glados` or `char:jarvis` from the same dropdown — each is a real voice plus a live effect chain. You can make your own, and downloading a character's "ideal base" voice (flagged in the voice library) sharpens it.
- **Make your own voice: Settings ▸ Runtime ▸ "VOICE / TTS"** — either **drop in a clip** (a short, clean ~6–20s .wav/.mp3/.mp4 of the voice you want) or click **"Record a voice now"** to capture 12 seconds from your mic on the spot. You can also say *"create a voice called Nova from clip.mp4"*. It sounds like your sample once the neural voice extra is installed (`pip install "eli-v2.0[natural]"`); until then it's saved and ELI uses a normal voice.
- **ELI's face reacts:** the little animated face shows how ELI feels, **moves its mouth while it speaks**, and shows a focused look while it's thinking.
- **Teach ELI your voice & tone:** "train my voice" — it learns your pitch/energy and even emotion cues, then adapts how it speaks to you.
- **If voice misbehaves:** "run voice diagnostics".

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12. Seeing your screen and images (vision)

[Everyone] ELI can actually **look**:

- **Your screen:** "what's on my screen?" · "what does this error say?" · "find the submit button"
- **Image files:** "describe cat.jpg" · "read the text in this image" (OCR)
- **Documents:** "summarise report.pdf" · "analyse data.csv" · "analyse the pdfs in that folder"
- **Continuous watching:** "watch my screen" (and "stop watching")

The first time you use vision, ELI may fetch a small vision model (you can pre-download it — §14). With a webcam set up, ELI even supports **eye-tracking control**: "enable gaze control", "calibrate gaze", then click where your eyes rest.

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13. The ELI server & using it from your phone

[Some setup] *Slightly technical, but only at setup.*

ELI includes a built-in **server** and **web app**. Start the server on your computer and chat with ELI from your **phone or tablet's browser** over your home Wi-Fi — while all the actual thinking stays on your computer.

```
flowchart TD
    A([Start the ELI Server<br/>on your computer]) --> B{Who should reach it?}
    B -->|Just this computer<br/>default| C[Private to localhost]
    B -->|My phone/tablet<br/>on home Wi-Fi| D[Start in LAN mode]
    D --> E[It prints an address<br/>+ a security token]
    E --> F[Open that address in<br/>your phone's browser]
```

```
F --> G[Enter the token once]
G --> H([Chat with ELI from your phone<br/>thinking stays on the PC])
C --> H
```

- Start it from the desktop app — **Settings → Web Server** (a "this computer" button and a "phone / Wi-Fi" button that shows the exact link to open). Or run `scripts/eli_serve.sh` (Mac/Linux) / `scripts/eli_serve.ps1` (Windows). Full details: `docs/SERVER_AND_WEB_APP.md`.
- The web app is **installable** — your phone's "Add to Home Screen" makes it open like an app, and the shell works offline. Light/dark themes; you can recolour the chat.

Starting the server, and what it prints

Start the server and it does two friendly things so you never have to hunt for the address: it **prints a clickable link** (Ctrl/Cmd-click it right in the terminal), and it **auto-opens your local browser** to the dashboard. (Set `ELI_API_NO_BROWSER=1` if you'd rather it didn't.)

- **This computer only (default):** the dashboard opens at `http://127.0.0.1:8081/`. No token — you're on the machine, so you're trusted automatically.
- **Phone / Wi-Fi (LAN mode):** it prints your computer's LAN address (e.g. `http://192.168.1.20:8081/`) with a **one-time token** baked into the link, plus a **QR code** in the Connect tab. Scan it or open the link and you're in.

The token, and the rotate button

The phone's token is **stable** — it's saved (permissions `0600`, under your config dir) and survives server restarts, so a paired phone stays paired; no re-scanning every reboot. Lost the phone, or just want a clean slate? Hit **rotate**: it mints a fresh token and instantly cuts off anything still using the old one.

How the link keeps your token safe

The token rides in the link's **fragment** — the part after the `#` (`…:8081/#token=…`). Browsers **never send the fragment to the server**, so your token can't land in the server's access log (or any proxy log in between). When the page loads it reads the token from the fragment, stores it locally, then **wipes it from the address bar** so it isn't left in your history either. Older links that used `?token=` still work — the page accepts both — but everything ELI prints or displays now uses the safe fragment form, on the desktop panel and both launcher scripts alike.

Voice from the phone needs HTTPS

Browsers block the microphone on a plain `http://` LAN address, so the phone **mic** needs a secure page. Start with `--https` (or the HTTPS toggle) and ELI runs a *second* server alongside on port **8443** with a self-signed certificate — accept the one-time "not private" warning and the mic works. Plain HTTP stays the primary path (it's what QR scanners open reliably); HTTPS runs *in addition*, purely for voice.

Switching the model from the dashboard

Settings → Model lists every model installed on your computer, with sizes. Pick one and ELI **hot-swaps** it live — no restart. Only real, installed files show up, so you can't point it at something that won't load, and only an **admin** can switch it.

Two servers, one machine

Worth knowing: "the server" is really **two** local services. The **web/API server** (port 8081) serves the dashboard, the phone app, and a small API. The **device server** is a separate MQTT service that talks to your smart-home gear (see the Home tab, below). Both run locally; both are yours. Everything about who's allowed in — tokens, roles, the loopback trust — is laid out in **Appendix C (Security)**.

Under the hood (for scripts & power users)

The web/API server is a **FastAPI** app served by **uvicorn** (`api/server.py`). You rarely touch it directly — the desktop app and the launcher scripts wire everything up — but if you script it, a handful of environment variables control it: - `ELI_API_HOST` — `127.0.0.1` (default, this-computer-only) or `0.0.0.0` (LAN mode). - `ELI_API_PORT` — the HTTP port (default `8081`); the HTTPS voice server uses `8443`. - `ELI_API_TOKEN` — set your own token; if unset, ELI uses the saved stable one (`config/api_token`, permissions `0600`) so a paired phone survives restarts. - `ELI_API_ALLOW_TOKENLESS` — honoured only on loopback; LAN mode always requires the token. - `ELI_API_NO_BROWSER=1` — start without auto-opening the local browser.

The launchers wrap all of this: `scripts/eli_serve.sh` (Mac/Linux) and `scripts/eli_serve.ps1` (Windows) start LAN mode, then print the phone link (with the safe `#token=` fragment) and the **exact** firewall command for your OS — the quickest way in from a terminal. The API itself is a small REST surface (chat, status/telemetry, model list/switch, voice, memory, research, audit); full endpoint detail lives in `docs/SERVER_AND_WEB_APP.md`.

If a phone can't connect

Nine times out of ten it's the firewall. When it starts in LAN mode ELI prints the **exact** command to open the port for your OS — run that, and make sure the phone is on the **same Wi-Fi**. It's all scoped to your local network; nothing is exposed to the internet.

What's in the web app

It has a sidebar with several sections: - **Overview** — a live dashboard (clock, GPU/CPU/RAM gauges, status, recent activity). - **Chat** — streamed replies with markdown/code, saved sessions, stop/regenerate, and voice in/out. - **Commands** — a searchable list of everything ELI can do. - **Home** — ELI's own **home-AI** (see below). - **System** — live, measured GPU/CPU/RAM/model telemetry. - **Research** — shared document corpora you can build and query **together**, with citations. - **Audit** — a tamper-evident log of every action. - **Admin** — (admin only) per-user activity, the risk-gate policy, and user management.

Roles (optional)

By default the person on the computer is the operator (full admin). If others share it, you can create accounts with roles — **viewer** (read-only dashboards), **member** (use everything), **admin** (also the Admin console) — in the Admin tab. Each gets a one-time link; the audit trail then attributes actions to the right person.

The Home tab — ELI as your home assistant

ELI controls smart devices **directly over MQTT** (ESPHome / Tasmota / Zigbee2MQTT) — no Home Assistant, no cloud: - **Find devices** on your network with one click (mDNS), or add one by its topics. - **Control** lights/switches, group them into **rooms**, and run room-wide on/off — from the app or by voice ("turn on the desk lamp", "turn off the kitchen"). - **Scenes** — save a set of device states as "Movie mode" and activate it (by voice too). - **Automations** — *when* something happens (a time, **sunrise/sunset**, or a device turning on/off), *do* something (control a device or run a scene), optionally *only if* a condition holds. - ELI **learns** how you use your home and **suggests** automations you can accept with one tap.

---

14. Models: picking, switching, and training your own

[Some setup] The **model** is the "brain" ELI thinks with — a file on your computer. Bigger = smarter but needs more graphics memory (VRAM). ELI picks a suitable one at install; change it anytime.

Which model should I get?

```
flowchart TD
    A([How much graphics memory<br/>VRAM do you have?]) --> B{Amount?}
    B -->|None / CPU only| C[qwen2.5-3b<br/>small and fast]
    B -->|~8 GB| D[qwen2.5-7b default<br/>or qwen3-8b]
    B -->|~12 GB| E[falcon3-10b<br/>or phi-4]
    B -->|24 GB+| F[qwen3.6-35b-a3b<br/>or falcon-h1-34b]
    C --> G([ELI auto-sizes it to your<br/>hardware — you tune nothing])
    D --> G
    E --> G
    F --> G
```

Downloading

- **Pick from a menu (choose any number of models):** `python -m eli.core.model_download --choose` — this is what the installer uses; you tick the ones you want (or `all` / `auto` / `none`).
- Let ELI choose one best-fit for your hardware: `python -m eli.core.model_download --auto`
- See the menu without downloading: `python -m eli.core.model_download --list`
- Grab a specific one: `python -m eli.core.model_download qwen2.5-7b`

You're not limited to one — download several and switch between them anytime (§"Switching models").

Command name	Model	Roughly needs
`qwen2.5-3b`	small & fast	4 GB graphics / or CPU
`qwen2.5-7b` *(default)*	great all-rounder	8 GB graphics
`qwen3-8b`	newer, big context, good reasoning	8 GB graphics
`falcon3-10b`	a step up	12 GB graphics
`phi-4`	strong reasoning for its size	12 GB graphics
`qwen3.6-35b-a3b`	very capable (mixture-of-experts)	24 GB / or slow on CPU
`falcon-h1-34b`	largest option	24 GB / or slow on CPU

You can also just **drop any `.gguf` file** into the `models/` folder and ELI finds it.

Switching models

In **Settings → Model** (or "Model menu / Auto Detect"), pick the file. ELI sizes it to your hardware automatically — you don't tune anything.

Using Ollama instead (Windows, macOS, Linux)

[Some setup] If you already run **Ollama**, ELI can use your Ollama models instead of a GGUF file — handy when you've curated models there, or want ELI to share one Ollama install.

1. Install and start Ollama (`https://ollama.com/download`). On Windows and macOS it starts itself; on Linux use `systemctl --user start ollama` (or `ollama serve`).
2. Pull at least one model, e.g. `ollama pull llama3.2`.
3. In ELI: **Settings → Model → Backend = Ollama**, then **Refresh Ollama Models** and pick one. It's also in the startup model picker and the toolbar dropdown.

Host and port. Leave *Host* blank/default for a normal local install. Set it if Ollama runs elsewhere — a different port (`localhost:9999`), another machine (`192.168.1.20`), or a container. You can type it any way that makes sense (`localhost:11434`, `127.0.0.1:11434`, a bare IP) — ELI fills in the missing pieces. **Ollama**

on another machine is supported: ELI treats a host you configured as a deliberate local service, so its offline-by-default guard won't block it and the Net toggle can stay off.

If ELI can't see your models: confirm Ollama is running (`ollama list` in a terminal). If it's on another machine, start it there with `OLLAMA_HOST=0.0.0.0:11434` so it accepts connections beyond that machine, and allow port `11434` through its firewall.

[Advanced] Training your own ELI-flavoured model (advanced, optional)

You can fine-tune a model so it speaks in **ELI's voice** out of the box.

See `blueprints/finetuning_guide.md` — a full step-by-step from choosing a base model, through extracting a voice dataset and training, to producing a ready-to-load `.gguf`. Crucially, ELI's **personality stays live** (never frozen into the model); training only shapes the *manner* of speech.

Training ELI on your own conversations

[Some setup] ELI can fine-tune a model on **the way you actually talk to it** — all on your computer, nothing uploaded. **Labs ▸ Training**, four steps.

1. Can this machine do it? ELI checks your graphics card, how much memory is free, and whether the training software is installed — then tells you plainly. If your card is too small it says how much it needed and what would fix it, rather than starting something that runs for days.

Works with NVIDIA and AMD cards. Apple and Intel graphics are detected but cannot run training; ELI says so instead of pretending.

2. What are you training? You need a **Hugging Face model folder**, which is a different thing from the `.gguf` file ELI chats with. ELI lists every one it can find on your computer. Pick one, give it a name, done.

3. Which conversations should it learn from? This is the important step. ELI gathers your past exchanges and sorts them for you:

- exchanges that were clearly broken are **removed automatically**;
- exchanges that look odd are **flagged for you to read**;
- the rest are shown as clean.

You approve the clean batch in one click and then read the flagged ones. **Nothing trains on anything you have not approved.** Typically that means reading a few hundred rather than every single exchange.

You can also edit an exchange — fix ELI's answer to what it *should* have said — and train on the corrected version.

4. Train. Pick Light (minutes), Standard, or Deep (hours). "Check readiness" runs every check without training. "Start training" runs it, showing progress and loss, and you can cancel.

> **After training finishes**, the result is a small "adapter" file, not a new brain. To actually chat with it, the adapter has to be merged into the base model and converted to `.gguf`. ELI tells you this when the run ends — it will not pretend the model has changed when it hasn't. See the fine-tuning guide for that step.

---

15. How hard ELI thinks (reasoning modes)

[Everyone] ELI has five effort levels. Faster modes answer quickly; deeper modes think longer and use more of ELI's internal helpers for hard problems.

Mode	Feel	Use it for
quick	snappy	greetings, simple commands, quick facts
normal	balanced (default)	everyday questions and tasks
advanced	more thorough	tricky, multi-step asks
research	digs deep	gathering and synthesising a lot
expert	maximum effort	the hardest problems

Ask "what reasoning mode are you in?" or "explain all your reasoning modes". For most people the default is right — ELI also deepens on its own when a problem clearly needs it.

---

16. Settings you'll actually touch

[Some setup] Most people never need these. The few that matter:

- **Model** — which brain ELI uses (§14).
- **Voice** — wake word, text-to-speech on/off, microphone.
- **Hardware / startup** — ELI auto-tunes; advanced users can nudge how much graphics memory to hold back, or how much context to use. Leave alone unless you know why.
- **Privacy** — ELI is offline by default; this is where anything network-related lives.

Two graphics cards? ELI can use both — set in the GPU profile settings; single-card just works.

---

17. Privacy & staying offline

[Everyone]

- **Nothing leaves your computer** unless you ask for something online (web search, news, weather, downloading a model). ELI tells you when it goes online.
- **Your data is yours** — conversations, memories, files, profile live in local databases. Delete anytime.
- **No accounts, no telemetry, no subscription.**
- A fresh install is a **blank slate** — ELI starts knowing nothing about you.

---

18. Troubleshooting

[Some setup] Most issues map to one of these. When in doubt, **ask ELI** — "run a full runtime audit" or "status" — it reports honestly.

"ELI is replying very slowly"

```
flowchart TD
    A([Replies feel slow]) --> B[Ask: 'what are you running on?']
    B --> C{Running on GPU<br/>or CPU?}
    C -->|CPU| D[Reinstall with graphics support,<br/>or pick a smaller model]
    C -->|GPU but big model| E[Switch to a smaller model<br/>see section 14]
    C -->|GPU, right-sized| F[That's normal speed for<br/>your hardware]
    D --> G([Faster])
    E --> G
```

Everything else

Problem	Try this
Replies look like gibberish	A model whose context is too small. Pick a recommended one (§14); ELI sizes context automatically.
“I couldn’t find a model”	`python -m eli.core.model_download --auto`
Memory/recall seems empty	Normal on a fresh install. Ensure the embedder downloaded: `python -m eli.core.model_download --aux`
Voice/wake word not responding	“run voice diagnostics”; check the microphone in Settings → Voice.
Won’t go online for news/search	By design (offline-first). The network is allowed per-request; check Settings → Privacy.
A habit didn’t run	Open **Habits** — confirm it’s enabled and the time is right; or “show my habits”.
Something seems broken	“run a full runtime audit” or “status”.

---

19. Quick phrasebook (cheat sheet)

Everyday - “open spotify and play lo-fi” · “pause” · “skip” - “what’s on my screen?” · “summarise report.pdf” - “set an alarm for 7am” · “set a timer for 10 minutes” - “what’s the news?” · “weather in Dublin” · “what time is it?”

Memory & you - “remember that …” · “what do you know about me?” · “my name is Alex”

Habits & proactive - “show my habits” · “yes” / “no” (to an offer) · “morning report” · “start proactive mode”

Writing & code - “write a report on X” · “write a script to …” · “fix the bugs in foo.py”

Voice - wake word “computer” + request · “change the wake word to athena” · “start dictation” · “say good morning” · “train my voice”

Check on ELI - “what are you running on?” · “status” · “what updates have you made recently?”

Models (in a terminal) - `python -m eli.core.model_download --list` · `python -m eli.core.model_download --auto`

---

20. Add-ons: plugins & the community marketplace

[Some setup] **Settings ▸ Marketplace.** Plugins add abilities to ELI. Some ship with it; others are written by **other people in the community**.

The honest bit, first

The marketplace belongs to the community, not to ELI’s author. Anyone can host one and anyone can publish to it. **Nobody vets what goes in**, so ELI will never tell you a plugin is safe — it isn’t in a position to know.

What it does instead, before a single file lands on your computer:

- **Checks the file is the file.** The listing publishes a fingerprint; ELI checks the download against it. If it doesn’t match, it refuses — that means the file was changed somewhere between the publisher and you.
- **Checks who wrote it**, if they signed it and you have chosen to trust their key. Unsigned isn’t blocked — it’s clearly labelled *unverified*.

- **Reads the code.** If it uses an ability it never declared, ELI refuses.
- **Scans it for malware** — eleven different checks, looking for things like remote access backdoors, password stealing, hidden start-up entries, disguised code, and crypto miners. If you have ClamAV or YARA installed, those run too.
- **Shows you everything it found**, then asks.

If a scanner couldn't run, ELI says the check was **partial**. It won't call something clean that it didn't fully check.

Permissions — like your phone

A newly installed plugin arrives **switched off with no permissions at all**.

When you enable it and it first wants to do something — reach the internet, read your files, use your microphone — ELI asks. The message names the plugin, says what it can do to you, and how risky that is. Four answers:

Answer	What happens
Allow once	Allowed now. You'll be asked again next time.
Always allow	Allowed from now on, no more asking.
Not now	Refused this time. You'll be asked again.
Never allow	Refused permanently. It can never ask again.

Every answer is recorded, and you can take any of them back in **Marketplace ▸ Permissions**.

If a plugin asks while nobody's there — an overnight scheduled job, or the phone server — the answer is automatically **no**. A plugin can't get permission by waiting until you're asleep.

Paid plugins

Some publishers charge. You buy from **them**, not from ELI, and paste the licence key in. ELI has no payment system, takes no cut, and **cannot confirm a purchase happened or that a refund is possible**. Treat it exactly like buying software from a stranger's website — because that's what it is.

MCP servers — one thing to know

MCP servers are add-ons that run as **their own separate program** alongside ELI. That means ELI's "stay offline" switch **cannot stop them** reaching the internet, and ELI can't see what they send. This isn't a flaw in ELI — it's how MCP works. Only add MCP servers you'd trust with that.

ELI does make sure they actually work: it checks the required software is installed, then genuinely starts the server and talks to it before saving anything. If it can't answer, it isn't added.

---

21. Making ELI your own: custom agents

[Advanced] An **agent** is a small specialist that runs alongside ELI's main reply — one that watches for a certain kind of request and contributes something extra.

You can write your own. A custom agent needs four things, and ELI won't accept it without them:

Objective	One sentence: what is this agent responsible for?
Instruction	What it should actually do — the prompt it runs on.
Triggers	When it should wake up (a keyword, a pattern, or a type of request).

How you'd know it worked	Checks ELI can run on the answer — must mention X, must not say Y, must be at least N characters, must be valid JSON.

That last one is the part people skip, and it's the reason it's required. Without it, an agent either seems fine or doesn't, and you can never tell which. With it, ELI **tests** the agent and an agent that fails its own test contributes nothing rather than quietly adding noise to your answers.

Agents are created switched off. You can edit one and reload without restarting ELI.

> If you write an agent as actual Python code rather than a specification, ELI treats it as untrusted until you approve it — and it scans the code first, recording what it found and when you approved it. Editing the file afterwards cancels the approval automatically.

---

22. Glossary

Term	Plain meaning
Model / GGUF	The "brain" file ELI thinks with. `.gguf` is just its file format.
VRAM	Memory on your graphics card. More VRAM → you can run a bigger, smarter model.
Embedder	A tiny helper model that powers ELI's memory/recall. Installed automatically.
Context	How much of the conversation ELI can "hold in mind" at once. ELI sets this for you.
Vision model	The model that lets ELI understand screens and images.
Wake word	The phrase that makes ELI start listening (default "computer").
Proactive mode	ELI doing helpful things on its own (briefings, suggestions).
Habit	A routine ELI learned and you approved, run on a schedule.
Fine-tuning	Optional advanced step to teach a model ELI's voice (§14).
Offline-by-default	ELI never uses the internet unless you ask for something that needs it.

---

23. Appendix A — Under the hood: how ELI thinks (the full pipeline)

[Advanced] *For the curious / technical. You never need this to use ELI — but here is the complete, verified path from your words to ELI's reply.*

Every request runs through one funnel — `CognitiveEngine.process()` — which takes one of three paths depending on what you asked and the reasoning mode. The diagram is the **real** pipeline (verified against the running code): a deterministic fast-path for plain commands, a **quick** path via the Agent Bus (the default), and a **deep** 12-stage Orchestrator for hard work.

```
flowchart TD
    U([Your input — text or voice]) --> R[Router<br/>regex priority pipeline<br/>+ LLM-intent fallback]
```

```
R --> P[CognitiveEngine.process<br/>action · args · confidence · mode]
P --> FP{Plain deterministic command?<br/>open app · media · volume · status · job}
FP -->|Yes · PHASE45 fast-path| FX[execute_action] --> VERB([Reply returned verbatim<br/>no LLM, instan
FP -->|No| MODE{Reasoning mode}

MODE -->|quick · default| BUS[Agent Bus dispatch]
BUS --> AG[15 specialist agents run over a DAG<br/>memory · knowledge-graph · system · file-code<br/>in
AG --> DR[DispatchResult<br/>grounding_confidence · agents_used · memory_context]
DR --> GE{Low grounding on a<br/>factual question?}
GE -->|Yes| ESC[Escalate: web tier / local tier / honest hedge] --> GOV
GE -->|No| KIND{What kind of result?}
KIND -->|self-contained action| GOV
KIND -->|grounded control| GSYN[Grounded synthesis] --> GOV
KIND -->|conversation / CHAT| SYS[Build enhanced system prompt<br/>persona + memory brief + context] -->

MODE -->|normal/advanced/research/expert · deep| ORCH[12-stage Orchestrator]

subgraph DEEP[The 12-stage Orchestrator]
  direction TB
  O1[1 · Intent routing] --> O2[2 · Persona lock]
  O2 --> O3[3 · HyDE query expansion]
  O3 --> O4[4 · Planner — channels + budgets for the mode]
  O4 --> O5[5-7 · Parallel retrieval<br/>keyword · FTS5 turns · FAISS vector · RAG · KG]
  O5 --> O8[8 · Hybrid merge]
  O8 --> O9[9 · Cross-encoder rerank]
  O9 --> O10[10 · Context assembly]
  O10 --> O105[10.5 · Persona handoff]
  O105 --> O11[11 · LLM generation]
end
ORCH --> O1
O11 --> O12{12 · Confidence vs threshold}
O12 -->|pass| GOV
O12 -->|too low| REPAIR[Repair / regenerate] --> GOV

GEN[LLM generation<br/>broker.infer / stream] --> GOV[Output governor<br/>+ No-Fake-Actions guard<br/>ne
GOV --> OUT([Reply to you])
VERB --> OUT
```

The retrieval stages (5–7) are where memory plugs in — that's Appendix B. The two guarantees baked into the end of every path: the **Output governor** cleans the reply, and the **No-Fake-Actions guard** ensures ELI never claims it did something it didn't.

A.1 — The router (before any path is chosen)

Your words first hit the **router**. It runs a **regex-first priority pipeline**: an ordered set of fast, exact checks (web realtime lookups, media controls, file/PDF, memory, OS control, …), each carefully shaped so it only fires on a real command — e.g. media controls need a whole word in command shape, so "metabolise" never triggers a media "tab". If nothing matches, an **LLM intent classifier** is the fallback. The router emits `{action, args, confidence}` and hands off to `CognitiveEngine.process()`. The router also remembers the *last used path* so follow-ups ("do it again", "the second one") resolve correctly.

A.2 — The three paths, and when each is taken

- **Fast-path (PHASE45) — no AI involved.** Plain, deterministic actions (open an app, pause music, set volume, report status, check a job) are executed directly and the result is returned **verbatim**. There's no model call, so these are instant and can't be "hallucinated". This is why "pause" never produces a chatty paragraph.

- **Quick path (the default) — the Agent Bus.** Conversational and lightly-grounded requests go to the **Agent Bus**, which runs up to **15 specialist agents** in parallel over a dependency graph, each contributing evidence (memory hits, knowledge-graph facts, system readings, code search, capability info, …). Their combined result carries a **grounding confidence** — an honest "is this actually backed by anything?" score.
- **Deep path — the 12-stage Orchestrator.** When you ask for depth (advanced/research/expert modes, or a hard question), ELI runs the full pipeline below: more retrieval, reranking, and a confidence check with automatic repair.

A.3 — The Agent Bus, in full

The bus **selects a relevant subset** of agents for your intent (or fans out broadly when unsure), runs them concurrently, and **aggregates** their findings into a `DispatchResult`. Key agents and what they bring:

Agent	Contributes
Memory	recalls facts/turns from SQLite + full-text + vector search
Knowledge-Graph	entities and how they relate (who/what/where)
System	runs executor actions (web search, audits, status)
File-Code	searches the actual codebase for grounded answers about ELI itself
Introspection	live runtime/self state
Capability	what ELI can do (the capability manifest)
Habit / Proactive / Reflection / Self-Improve	routines, signals, insights, fixes
Plugin / Voice / Frontier / Orchestrator	plugin dispatch, speech state, deeper reasoning hooks, planning

After the bus returns, a **grounding-escalation hook** checks: *is this a factual question that came back weakly supported?* If so, ELI escalates — a web tier (if you allow online), a local tier, or an **honest hedge** ("I'm not certain, but…") rather than a confident guess. Then the result is finalised: a self-contained action returns directly, a grounded control action gets a short grounded synthesis, and a conversation goes to generation with the full persona + memory context.

A.4 — The 12 stages, one by one (the deep path)

1. **Intent routing** — settle exactly what you're asking for.
2. **Persona lock** — fix ELI's identity/voice for this turn so it stays consistent.
3. **HyDE query expansion** — ELI drafts a *hypothetical answer* and searches with **that** too, which dramatically improves what memory retrieves (you find better matches by searching with an answer-shaped query, not just the bare question). Skipped for very short queries.
4. **Planner** — decide *which* retrieval channels to use and how big the budgets are, tuned to the reasoning mode (fast / balanced / deep). 5–7. **Parallel retrieval** — several searches run at once: keyword, **full-text (FTS5)** over your conversation history, **FAISS vector** (semantic) search, optional RAG, and the **knowledge graph**. (This is the seam where Appendix B's memory system feeds in.)
5. **Hybrid merge** — combine the hits from every channel into one candidate set.
6. **Cross-encoder rerank** — a smarter model re-scores the merged candidates for true relevance to your question, and the best rise to the top. (Skipped for non-conversational asks.)
7. **Context assembly** — build the grounded context block that will inform the answer. 10.5. **Persona handoff** — wrap that context with ELI's live persona brief for generation.
8. **LLM generation** — the model writes the reply (streaming or one-shot). For the deepest modes this is a two-part private-reasoning-then-condense step.
9. **Confidence check** — score the answer against a threshold; if it's too low, ELI **repairs/regenerates** rather than shipping a weak answer.

A.5 — The two end-of-path guarantees

Every path, no matter which, ends at: - **Output governor** — normalises and cleans the reply (strips artefacts, enforces formatting). - **No-Fake-Actions guard** — if ELI's reply *claims* it did something (played a song, fixed a file), the system verifies it actually happened; if not, the claim is replaced with the honest truth. This is enforced in code, not left to the model's goodwill.

---

24. Appendix B — Under the hood: how ELI remembers (the full memory system)

[Advanced] *For the curious / technical. ELI's memory is a genuine hybrid — relational + full-text + vector + graph — all local. This is the complete input → storage → recall path (verified against the code).*

```
flowchart TD
    subgraph IN[INPUT — everything ELI may write]
      direction LR
      I1[Conversation turns]
      I2["'Remember X' durable facts"]
      I3[Observations + user patterns]
      I4[Habits + habit events]
      I5[Failures · corrections · improvements]
      I6[Knowledge-graph entities + relations]
      I7[User Model brief]
    end
    IN --> MEM[Memory class<br/>memory.py routes every write]

    MEM --> D1[(user.sqlite3<br/>memories · turns · habits<br/>patterns · user_model · KG)]
    MEM --> D2[(agent.sqlite3<br/>dispatch + metrics)]
    MEM --> D3[(system_index.sqlite3<br/>apps · files · dirs)]
    MEM --> D4[(coding_memory.sqlite3<br/>bug fixes)]
    MEM -->|embeds new text| V[[FAISS vector index<br/>nomic embedder<br/>serialized — segfault-safe]]
    MEM -->|indexes text| F[[FTS5 full-text<br/>memories + turns + KG]]
    MEM -->|entities/relations| KG[[Knowledge graph<br/>subject - predicate - object]]

    Q([Recall request<br/>from pipeline stages 5-7]) --> RC[recall_memory<br/>hybrid retriever]
    V --> RC
    F --> RC
    RC --> S1[1 · FAISS vector search — primary]
    S1 --> S2{Cold index or<br/>fewer than limit/2 hits?}
    S2 -->|Yes| S3[2 · FTS5 / LIKE keyword supplement]
    S2 -->|No| S4
    S3 --> S4[3 · Noise filter<br/>drop ELI's own insights / reflections /<br/>session summaries / rows over
    S4 --> S5[4 · Importance-weighted ordering<br/>COALESCE importance 0.5]
    S5 --> CTX([Grounded context handed to the prompt])
    KG -.->|context_for_prompt| CTX
    I7 -.->|read fresh every turn| CTX

    subgraph MECH[UPKEEP MECHANISMS]
      direction LR
      M1[Weight decay<br/>fades by AGE, half-life stretched by importance]
      M2[Embedder lock<br/>one-at-a-time embedding]
      M3[recall_log<br/>records what was recalled]
    end
    M1 -.-> D1
```

Why it's trustworthy: the **noise filter** stops ELI's own past replies and reflections from resurfacing as if they were *your* facts; the **embedder lock** keeps the vector engine from crashing under concurrent use; and **weight decay** is the gentle "forgetting" that lets stale topics fade while active ones stay fresh — a memory fades with its age, and every time one is recalled its half-life stretches, so what you keep coming back to keeps its place. Nothing is ever deleted by fading, and **consolidation** folds duplicate entries into one rather than letting the same thought accumulate. Nothing here leaves your computer.

B.1 — What ELI writes (the inputs)

Everything that flows into memory passes through one **Memory** module, which decides where each item belongs: - **Conversation turns** — what you said and what ELI replied, with timestamps. - **Durable facts** — anything you explicitly ask it to "remember". - **Observations & user patterns** — quiet signals about your interests, focus, and tone that build the living profile. - **Habits & habit events** — your repeated actions and the routines learned from them. - **Failures, corrections, improvements** — ELI's own mistakes and fixes, kept for self-learning (these are deliberately **excluded** from normal recall — see the noise filter). - **Knowledge-graph entities & relations** — named things and how they connect. - **User Model brief** — a compact, pre-rendered summary of who you are, refreshed over time.

B.2 — Where it's stored (four local databases + three indexes)

Everything is plain local files on your machine — no server, no cloud: - **`user.sqlite3`** — the main store: memories, conversation turns, habits, patterns, the User Model, and the knowledge graph. - **`agent.sqlite3`** — ELI's internal agent activity and performance metrics. - **`system_index.sqlite3`** — an index of your apps, executables, files, and folders (so "open the file manager" is instant). - **`coding_memory.sqlite3`** — remembered code bug-fixes.

On top of the relational data sit three search indexes: - **FAISS vector index** — every stored text is turned into an "embedding" (a list of numbers capturing meaning) by a small local **nomic embedder**, so ELI can find things by *meaning*, not just exact words. Embedding is done **one-at-a-time** behind a lock (the embedder isn't thread-safe; serialising it prevents crashes). - **FTS5 full-text index** — fast exact/keyword search over memories, conversation turns, and the knowledge graph. - **Knowledge graph** — a subject → predicate → object web (e.g. *you* → *own* → *a Tesla*) that ELI can walk to answer relational questions.

B.3 — How recall works (step by step)

When the cognition pipeline (stages 5–7) needs memory, it calls **`recall_memory`**, a genuine hybrid retriever that runs in this order: 1. **Vector search first (FAISS).** The semantic index is the primary path — it finds things that *mean* the same even if worded differently. 2. **Keyword supplement (FTS5/LIKE) — only if needed.** If the vector index is cold/empty or returns fewer than half the requested results (e.g. a very short query), a keyword search tops it up. (When another part of the system already did the vector search, this step is skipped to avoid double-searching.) 3. **Noise filter.** Results are scrubbed of ELI's *own* output — assistant insights, reflections, session summaries, the "orchestrator" source, and any over-long blobs (>1500 chars). This is the safeguard that stops ELI's past musings from masquerading as *your* facts. 4. **Importance-weighted ordering.** What survives is ranked by an importance score so the most significant memories surface first.

Alongside this, the **knowledge graph** contributes a lightweight relational context, and your **User Model brief** is read **fresh every single turn** — that's why ELI stays in context about who you are without you repeating yourself.

B.4 — Upkeep (the mechanisms that keep memory healthy)

- **Weight decay** — old, low-importance memories are gently faded (multiplied down over time), so abandoned topics naturally recede while active ones stay prominent. This is ELI's "forgetting".
- **Embedder serialization** — the single lock around the embedder that keeps the vector engine stable under concurrent access.

- **Recall log** — a record of what was retrieved, used for tuning and for honestly explaining *where* a piece of knowledge came from ("how do you know that?").

B.5 — Privacy, restated

Every database and index in this appendix is a local file under your user profile. There is no sync, no upload, no account. Delete the files (or use the **Memory** tab) and that knowledge is gone. A fresh install starts with the full structure but **zero** personal rows — a true blank slate.

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25. Appendix C — Under the hood: how ELI stays safe and yours (security)

[Advanced] *For the curious / technical. ELI is meant to be safe by construction, not by good intentions — every guard below is enforced in code, not left to a setting you might forget to flip. All of this is verified against the running codebase.*

C.1 — Offline by default (the socket failsafe)

ELI doesn't just *prefer* to stay offline — it's physically stopped from reaching the internet unless you asked for something that needs it. At startup, `netguard` installs a **process-wide socket guard**: it patches the low-level `connect()` call so that *any* outbound connection to a non-loopback address raises an `OfflineError` while ELI is offline. Loopback and local sockets (the phone web server, the local model) always work. So even if some stray bit of code *tried* to phone home, the operating-system call itself is blocked — belt and braces. When you do turn networking on for a task, that flip is written to the audit ledger; "internet on" is never silent.

Ask for something online while offline and you don't get a crash or an invented answer — you get an honest refusal: *"I can't fetch that right now, I'm offline."*

C.2 — The fail-closed command gate

ELI can run shell commands and open apps, so that path is gated hard. Before any command runs it's checked against an **allowlist** by the SecurityManager. The part that matters: if the security layer can't load for any reason, the check **fails closed** — it returns *denied*, not *allowed*. A broken guard blocks the action; it never quietly waves one through. Sandboxed runs are limited to a fixed set of allowed executables.

C.3 — The approval gate (nothing risky happens behind your back)

Anything ELI proposes on its own — an autonomous action, a self-improvement patch, a scheduled job — goes through the **approval engine** before it can execute. Each proposal carries an *action class*, and the engine decides: auto-approve the harmless ones, or hold the riskier ones until **you** confirm. Who is even allowed to propose what is also gated — a background daemon can't quietly emit a dangerous action. There's a **Full Control** mode that lifts the confirmation barrier if you want ELI to just act, but that's your explicit choice, and it's off by default.

C.4 — The phone: who's allowed in

The web server (§13) is locked down. On the same machine (loopback) you're trusted automatically. From the network, every request must carry a **bearer token** — and that token is never stored in plain text, only its SHA-256 hash. The token travels in the link's URL **fragment** (after the #), which browsers never transmit to the server, so it can't appear in the access log; the page reads it locally and immediately clears it from the address bar. Define one or more users and **RBAC** switches on: each token maps to a role (admin / member), and admin-only actions — changing settings, switching the model — reject anyone who isn't admin. With no users defined it's single-operator mode: just you. The **rotate** button issues a fresh token and kills the old one, so a paired phone can be cut off in one tap.

C.5 — The audit ledger

Security-relevant events — settings changes, the network being turned on or off, research collaborations — are written to a **tamper-evident audit ledger**: a local, append-only trail so there's always an honest record of what changed and when.

C.6 — Privacy, restated

Every guard, token, ledger, and database above is a local file under your own user profile. No account, no sync, no telemetry. ELI's default posture is *closed* — offline, gated, and yours. You open doors deliberately, one at a time.

That's the whole assistant — inside and out. You don't need any of the last three appendices to use ELI; just talk to it in plain English, and ask "what can you do?" whenever you're curious. Everything here happens on your computer, for you, privately.